WHAT THE CELL · DP BIOLOGY · WORKSHEET · SL AND HL

C3.2 Defence against disease

NAME	DATE	TOTAL
		_____ / 70

About 70 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make. Look at the bold command term: "state" needs a brief answer, while "explain" needs reasons.

Section A: Recall

10 MARKS

1 **OUTLINE** two structural differences between a bacterium and a virus. [C3.2.1] [2]

2 **STATE** one physical defence and one chemical defence provided by the skin. [C3.2.2] [2]

3 **STATE** the type of white blood cell in the innate immune system that engulfs pathogens. [C3.2.4] [1]

4 **DISTINGUISH** between an antigen and an antibody. [C3.2.7] [2]

5 **STATE** what is meant by immunity. [C3.2.10] [1]

6 **STATE** two ways in which HIV can be transmitted. [C3.2.11] [2]

Section B: Understanding

17 MARKS

7 **OUTLINE** how a blood clot forms once clotting factors have been released. [C3.2.3] [3]

8 **DESCRIBE** how a phagocyte reaches and destroys a bacterium. [C3.2.5] [4]

9 **EXPLAIN** how a B-cell is activated and how the body then produces enough antibody to fight an infection. [C3.2.6, 8, 9] [4]

10 [C3.2.13]

a **EXPLAIN** why an antibiotic can kill bacteria in a patient without harming the patient's own cells. [2]

b **EXPLAIN** why antibiotics do not control viral infections. [1]

11 **COMPARE AND CONTRAST** the innate and adaptive immune systems. [C3.2.4] [3]

INNATE	ADAPTIVE

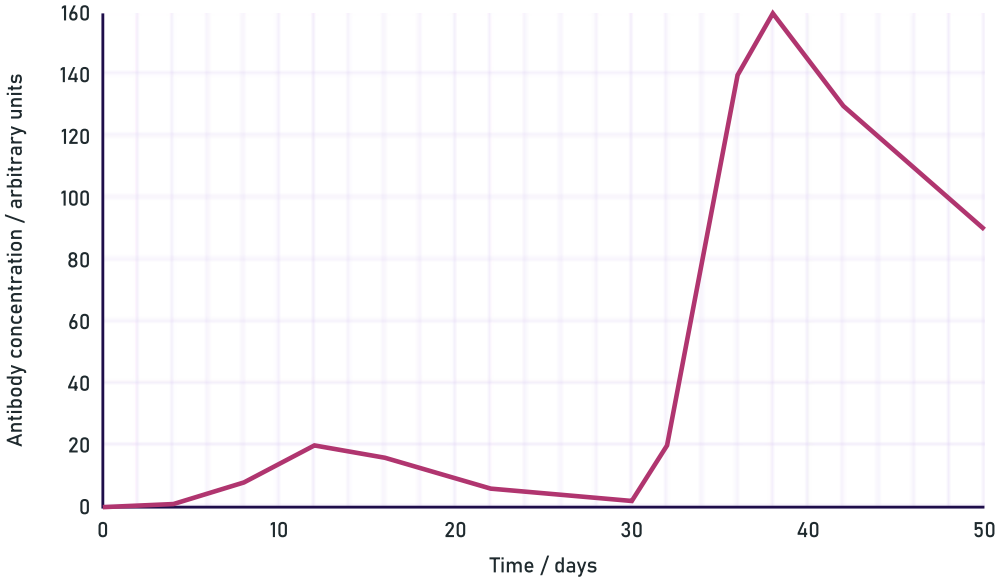
Section C: Working with data

34 MARKS

All data in this section are invented for practice. The patterns are realistic, but they do not describe real people or places.

12 [C3.2.10]

A person was exposed to a pathogen on day 0 and to the same pathogen again on day 30. The graph shows the concentration of the specific antibody in their blood.



a STATE the peak antibody concentration after the first exposure. [1]

b STATE how many days it took to reach the peak after each exposure. [1]

c CALCULATE the percentage change from the first peak to the second peak. [2]

d EXPLAIN the difference between the two responses. [3]

13 [C3.2.14]

A hospital laboratory recorded the percentage of *Staphylococcus aureus* samples that were resistant to one antibiotic.

YEAR	2010	2012	2014	2016	2018	2020
RESISTANT SAMPLES / %	8	11	16	21	27	30

a DESCRIBE the trend in the data. [1]

b CALCULATE the percentage change in resistant samples from 2010 to 2020. [2]

c EXPLAIN how natural selection could produce this rise. [4]

d SUGGEST two actions that could slow this rise. [2]

14 [C3.2.17]

The table shows measles vaccination coverage and measles cases in a fictional country.

YEAR	2000	2005	2010	2015	2020
VACCINATION COVERAGE / %	60	75	88	94	95
CASES PER 100 000 PEOPLE	420	150	35	6	4

a DESCRIBE the relationship between coverage and cases. [1]

b CALCULATE the percentage change in cases from 2000 to 2015. [2]

c EXPLAIN why people who were not vaccinated also became less likely to catch measles. [2]

d EVALUATE the claim that these data prove vaccination caused the fall in cases. [2]

15 [C3.2.18]

The table shows weekly COVID-19 cases per 100 000 people in two fictional countries.

	START OF MONTH	END OF MONTH
COUNTRY A	300	240
COUNTRY B	not given	160

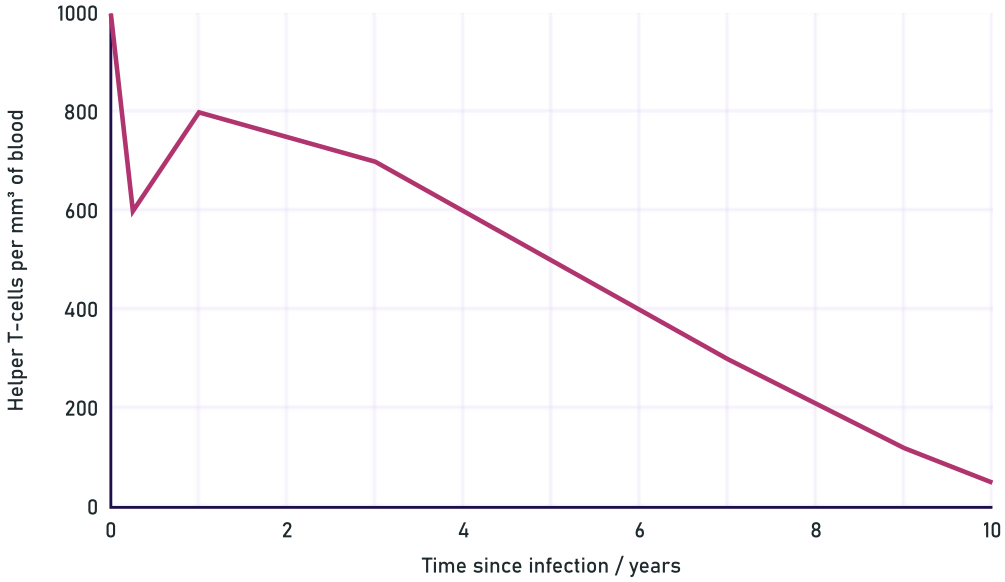
a **CALCULATE** the percentage change in cases in Country A during the month. [2]

b **CALCULATE** the percentage difference between Countries A and B at the end of the month. [2]

c **SUGGEST** two reasons, other than the spread of the virus, why the reported figures might differ. [2]

16 [C3.2.12]

The graph shows the helper T-cell count of a person with untreated HIV infection.



a **STATE** the helper T-cell count 5 years after infection. [1]

b **DEDUCE** when the count first falls below 200 cells per mm^3. [1]

c **EXPLAIN** why this fall makes the person more likely to develop opportunistic infections. [3]

Section D: Longer answers

9 MARKS

17 **OUTLINE** how two named zoonoses reach humans, including the type of pathogen and the animal source of each. [C3.2.15] [4]

18 **EXPLAIN** how a vaccine gives long-term immunity without causing the disease. [C3.2.16] [5]

Original questions written for What the Cell. Textbook reference: Biology for the IB Diploma (Hodder, 2023), pp. 517–547. Not an official IB resource.